

NATIONAL OPEN DATA INNOVATION CHALLENGE 2 — ENERGY & ENVIRONMENT

GridIQ — Solutions Report

Team Eukaryote1.0 | Gaborone, Botswana | September 2026

Plan the grid as one system, not three separate bets.

Anchor: 642 GWh of electricity was lost in FY2023 — 14.51% of units — by BPC's own accounting.

Executive Summary

Botswana plans its electricity system as a series of separate projects. Generation, storage and grid are decided one at a time, so the country stays over-reliant on coal at Morupule and on imported power, rural villages wait for a last mile that is never the cheapest option, and the network is run without knowing which assets matter most.

GridIQ is one shared, data-grounded model of that system, with four engines reading the same topology, demand and injection points:

1. **System-wide optimisation** — what to build, where, how much and how to connect, against one objective.
2. **Flow, congestion & losses** — where constraints bind and what they cost.
3. **Asset-risk screening** — which assets are structurally critical, plus the data contract that would unlock prediction.
4. **VillageFit** — the right technology for each unconnected village, scored against the cost of a line.

Because all four read the same substrate, no recommendation can silently contradict another. This is what makes GridIQ a platform rather than four scripts.

Our differentiator is that the model's loss output is **checkable against a figure BPC has already published** (642 GWh / 14.51% in 2023; 793 GWh / 16.62% in 2024). So the demo is a falsifiable test, not a presentation: if the numbers reconcile, every other engine inherits the credibility of a substrate that just proved itself. All four engines are implemented in the working prototype; the current focus is verification and closing the remaining data gaps.

One-sentence pitch: GridIQ helps BPC and Ministry planners — and the households and villages they serve — who struggle with a grid planned project-by-project, by providing one shared model of generation, storage, grid and demand, so that supply is planned at the lowest system cost, losses fall, and the last mile is served by the right technology.

1. Problem Definition & Context

1.1 Supply is planned project-by-project

Botswana's fleet is dominated by two coal stations — Morupule A (132 MW) and Morupule B (600 MW) — plus diesel peaking. Installed capacity sits around 892 MW against peak demand of 678 MW (2024), yet on a five-year rolling average domestic generation has covered only about 62% of supply. The rest is imported, and in Q1 2026 80.9% of imports came from Eskom alone. Local generation has covered as little as ~44% of distribution in a quarter. Meanwhile, new projects are appraised on resource potential alone: a strong solar site far from load can force a costly line upgrade the resource map never shows.

Who is affected	How they are harmed
Households and businesses BPC and Ministry planners	Unreliable supply and import-priced electricity. Must choose what to build next with no system-cost view.
Budget and tariff payers	Grid spend that a better-placed project would not have needed.

Failure mode	Consequence	Scale
One dominant generation site	A single coal fleet's availability sets national supply.	National
One dominant import corridor	Cost and reliability exposed to one supplier.	National
Project-by-project appraisal	Stranded grid spend; sub-optimal siting.	Every new project

1.2 The network is operated reactively

There is limited published visibility of where the network is congested, or which assets have no redundant path. Faults are answered after they happen. Public asset age, condition, inspection and failure data for BPC and other Botswana utilities is limited: no open dataset was found.

Who is affected	How they are harmed
Households and businesses BPC crews Planners	Outages they cannot anticipate or avoid. Respond to failures instead of pre-empting them. Cannot rank where maintenance or redundancy pays.

Failure mode	Consequence	Scale
No criticality view	A non-redundant asset can island a load pocket.	Network-wide
No condition data	Maintenance is reactive, not prioritised.	Every asset

1.3 The last mile is stuck

As of October 2024, 463 of Botswana's 565 gazetted villages were connected — 102 still unconnected, roughly one in six. These are exactly the low-density, far-flung settlements skipped in earlier phases, where line extension is disproportionately expensive. Government's stated fix is solar off-grid electrification, applied as the default rather than chosen per village.

Who is affected	How they are harmed
Rural households	No grid power; limited study, enterprise and health services.
Rural students The state budget	A government laptop with nowhere to charge it. Pays for a generic rollout instead of the cheapest fit.

Failure mode	Consequence	Scale
One technology prescribed everywhere	Poor fit where sun is weak and wind/biomass is strong.	~100 villages
No per-village comparison	Line extension assumed, not tested against mini-grids.	~100 villages

2. Solution Description

2.1 The shared substrate

GridIQ builds one network model of Botswana: buses (substations, generation, import gateways, load nodes) and edges (400/220/132 kV lines), with district demand, access gaps and resource potential attached. Every engine reads this same substrate, so a siting recommendation from Engine 1 can be re-priced by Engine 2 and re-located by Engine 4 without contradiction. *(The topology is drawn from OpenStreetMap/OpenInfraMap — the closest open substitute for BPC’s own grid data — cross-checked against the World Bank/AICD Botswana grid layer (2006-vintage, which captures only ~40% of the current 400/220/132 kV network), and patched with BPC’s publicly listed additions (North West Phase 1, Mochudi–Phakalane 132 kV, Mochudi and Ramotswa and Tlokweng substations)).*

2.2 Engine 1 — System-wide optimisation (implemented: screening scope)

- **What it is:** an annual least-cost capacity-expansion LP, plus a 24-hour benchmark chronology.
- **What it computes:** the mix of solar, wind, storage and imports for a given year that meets peak and reserve at least cost — imports plus capital plus losses — rather than judging one project at a time.
- **Inputs:** observed annual generation/import balance, SAPP peak and reserve anchors, published renewable/storage cost benchmarks, and user scenario limits.
- **User-facing artifact:** a ranked build mix for the year, plus a representative-day dispatch chronology. It is deliberately not a chronological production-cost model, and says so in the output.

2.3 Engine 2 — Flow, congestion & losses (implemented: benchmark scope)

- **What it is:** a linearised DC power-flow screen over the mapped topology — a single-transfer study and a balanced multi-injection system study.
- **What it computes:** branch flows, utilisation and I2R loss sensitivity from explicit injections. *(Power is not “routed”: current divides across all parallel paths by impedance, so we model constraints, not shortest paths.)*
- **Inputs:** mapped 132/220/400 kV geometry, standard-type R/X/current envelopes in three sensitivity cases, and explicit injections.
- **User-facing artifact:** a flow/utilisation view and a loss-reconciliation panel against BPC’s published series — **2023 anchor 642 GWh / 14.51%, 2024 793 GWh / 16.62%** (now sourced to BPC IR 2024/25) — cross-checked by AFREC (~630 GWh / 13.8%, 2023) and the World Bank (15.35%, 2021). Scope is stated openly: benchmark model, not BPC state; transmission-only versus BPC’s T&D. Our independent transmission-loss engine is kept as a separate cross-check at `GET /api/engines/loss-reconciliation`.

2.4 Engine 3 — Asset-risk screening, with a data contract (implemented)

- **What it is:** topology-based structural criticality, not a prediction model.
- **What it computes:** bridges, articulation points, N-1 geometry islanding and edge betweenness, ranked into a maintenance/redundancy watch-list.
- **Inputs:** mapped topology alone.

- **User-facing artifact:** a ranked watch-list plus the **BPC engineering data contract** (docs/data_contract) — the exact fields (bus/branch IDs, ratings, R/X, switch state, time-aligned loads, condition and failure history) that would upgrade the proxy into a validated model.

2.5 Engine 4 — VillageFit (implemented)

- **What it is:** a per-settlement technology recommender.
- **What it computes:** the best-fit mix of solar, wind, biogas, biomass or hybrid — with storage — scored against the cost of extending the line, including how the village uses power (e.g. evening study, where solar alone fails). Biogas is computed only when an explicit collectable manure mass is supplied.
- **Inputs:** a pilot-village register or custom settlement, live NASA POWER resource context, World Bank GEP/DRE planning indicators, distance to mapped lines, and benchmark community costs.
- **User-facing artifact:** a right-sized recommendation with the conditions that would change it; the “no-build” alternative feeds back into Engine 1.

Integration note: VillageFit is activated from the same substrate as the national engines. A village’s power is not a separate rural project; it is a system-cost decision the national plan currently makes project-by-project — and GridIQ makes it comparable.

2.6 The decision GridIQ is designed to change

These are reasoned inferences from the data and policy context, not measured findings.

Today, BPC’s transmission-planning engineer must decide whether a proposed solar plant can connect to an existing corridor, using resource maps and manual experience, but there is **no public view** of network headroom or of the system-cost consequence of one connection option versus another. (*Grounding: the OIC2 hub names the grid planner as this track’s target user, and BPC grid and load data are not openly available.*)

GridIQ combines the transmission topology, district demand and resource layers to **estimate** the binding constraint and the redispatch/loss cost of each option, under stated line-rating assumptions.

This allows the decision to be made on system cost, not resource potential.

Today, a student in an unconnected village like Gomolemo must decide how to study in the evening, and there is **no published comparison** of whether his village is better served by a line or a mini-grid. **GridIQ’s VillageFit** scores each unconnected village’s own resource mix against line-extension cost. (*Representative scenario based on documented evidence — 102 unconnected villages, 14.3% access in Ngwaketse West — not a documented individual; the laptop detail is illustrative.*)

3. Data & Technical Explanation

3.1 Data sources

Taxonomy: **REAL** = confirmed in the OIC2 hub; **EXTERNAL** = open Botswana/utility source outside the hub, cited directly; **BENCHMARK** = regional or methodological analogue, not Botswana data; **DEMANDED** = not openly available as of Sept 2026 (we call for its release); **ASSUMED** = model assumption, labelled.

Dataset	Source	Type
Electricity generation & distribution by source, quarterly 2016–2026	Statistics Botswana	REAL

Dataset	Source	Type
National Energy Use Survey 2022/23 (district demand, access)	Statistics Botswana	REAL
Africa Energy Balance 2025 (plant efficiency, losses)	AFREC, African Union	REAL
OpenInfraMap & Global Power Plant Database	OpenInfraMap / WRI	REAL
Global Solar Atlas, Global Wind Atlas, NASA POWER	World Bank/DTU; NASA	REAL
SAPP load & interconnection-capacity reports	SAPP	REAL
2022 Census, sub-district shapefile, Meta population density, OSM/HDX extracts	Statistics Botswana / Meta / OSM	REAL
Agricultural Census 2025 (livestock), FAO livestock production	Statistics Botswana / FAO	REAL
BPC Integrated Annual Report 2023 (T&D losses)	BPC	EXTERNAL
Botswana Electricity Transmission Network (2006, cross-check)	World Bank / energydata.info	EXTERNAL
Botswana techno-economic dataset (T&D and plant costs)	Zenodo / Loughborough CCG	EXTERNAL
Eskom Data Portal (import-side benchmark)	Eskom	EXTERNAL
Africa Electricity Transmission & Distribution Grid Map (2017)	World Bank / ESMAP	BENCHMARK
gridfinder predicted global grid	World Bank / ESMAP	BENCHMARK
SciGRID / GridKit / POMATO line parameters (r, x by voltage)	Open research	BENCHMARK
Global Electrification Platform / OnSSET	World Bank / ESMAP	BENCHMARK
ENTSO-E Transparency Platform / AEMO	Europe / Australia	BENCHMARK
Grid infrastructure & demand data (ratings, loads)	BPC	DEMANDED
Grid generation statistics & average demand	BOCRA	DEMANDED
Business register: large industrial consumers	Statistics Botswana	DEMANDED
Line ratings/impedances	Benchmarked from SciGRID/POMATO defaults	ASSUMED

3.2 Regional & benchmark sources

Where Botswana does not publish a layer, we use labelled analogues rather than leaving a blank — never presenting them as Botswana data. The **Africa grid map** (World Bank/ESMAP) cross-checks our OSM topology and adds cross-border corridors. **SciGRID** / **GridKit** / **POMATO** provide default per-km resistance and reactance by voltage class, so assumed line parameters are calibrated rather than guessed. The **Global Electrification Platform** / **OnSSET** supplies the least-cost

electrification method for Engine 4. **ENTSO-E** and **AEMO** are methodological analogues for load shapes, congestion and loss factors. For context: other countries do publish full open grid views — the EU mandates it (ENTSO-E), and the US and Australia publish transmission and constraint data — so Botswana’s gap is one of publication, not possibility.

3.3 Technology stack

Implemented today (v1.7).

Layer	Technology	Role
Runtime	Python + FastAPI/Uvicorn	API and static server.
Fallback runtime	Dependency-free stdlib server	Same core API contract if FastAPI cannot be installed.
Evidence substrate	Bundled JSON + SHA-256 manifests	Observed/derived/benchmark/reference separation and traceability.
Engine 1	scipy linprog (HiGHS)	Annual least-cost capacity-expansion screen.
Engine 2	scipy sparse DC power flow	Single-transfer and balanced multi-injection benchmark studies + I2R post-processing.
Engine 3	networkx (+ pure-Python fallback)	Bridges, articulation points, N-1 islanding, edge betweenness.
Engine 4	Python + bundled benchmark/cost tables	VillageFit settlement screening.
API surface	FastAPI routes under <code>/api/e1</code> , <code>/api/e2</code> , <code>/api/e3</code> , <code>/api/e4</code> , <code>/api/engines</code>	Planning engines and the independent loss cross-check.
Frontend	HTML/CSS/vanilla JS control centre (Leaflet + Chart.js)	Overview, Network, Optimizer, Criticality, VillageFit, Generation, Storage, Demand, Data, Audit.
Live connectors	OSM/Overpass, NASA POWER, WorldPop, World Bank GEP/DRE, Eskom benchmark chronology	External evidence with degraded states, never substitutes.

Target architecture (planned, not implemented). PostgreSQL + PostGIS + pgRouting; Celery/Redis workers; vector tiles + COG/TiTiler rasters; Next.js + MapLibre/Deck.GL. This is the scale-out path, not the current build — the report does not present it as delivered.

3.4 How people use GridIQ

GridIQ is used through a control-centre UI. Three journeys matter most, and each runs end to end on the implemented prototype.

Planner flow — decide the next connection (Engines 1 + 2 + 3).

1. The planner opens **Overview** and reads the national position: generation/import mix, published T&D losses (BPC 2023 anchor 642 GWh / 14.51%; 2024 793 GWh / 16.62%) and SAPP peak/reserve anchors.
2. On **Network**, they load the mapped grid (`/api/osm/power`) and see voltage classes, topology and evidence labels — no BPC operating state is implied.
3. They run a **benchmark system study** (`/api/e2/system-benchmark`) with explicit injections to see branch flows, utilisation and loss sensitivity, then `POST /api/e2/loss-reconciliation` to compare a modelled loss against BPC’s published series.

4. On **Optimizer**, they run the least-cost screen (`/api/e1/optimize`) for the year, and optionally the representative-day chronology (`/api/e1/representative-day`), to get the generation/storage/import mix.
5. They test a candidate site (`/api/e1/site-screen`) to see the grid-connection cost and whether a smaller local option wins.
6. They open **Criticality** (`/api/e3/criticality`) for the bridges/N-1/betweenness watch-list, and take the **BPC engineering data contract** as the concrete ask that would upgrade the screen to operational validation.

Community flow — power one village (Engine 4, VillageFit).

1. On **VillageFit**, the user selects a verified pilot village or enters a custom settlement (name, lat/lon, population).
2. GridIQ pulls live NASA POWER resource context and World Bank GEP/DRE planning indicators.
3. It scores solar, wind, biogas and hybrid against the cost of extending the line — biogas only when an explicit collectable manure mass is supplied.
4. It returns a right-sized recommendation and the conditions that would change it; the “no-build” alternative feeds back into the national plan (Engine 1).

Custodian flow — verify any number (Data + Audit).

1. On **Data**, the user sees the dataset catalog and the **benchmark gap-resolution register** — which gap each benchmark closes, and what it must never claim.
2. On **Audit**, they see scores computed from named evidence gates, with **BPC operational validation held at 0.0**.
3. Every figure traces to its source and its SHA-256 manifest entry; unavailable fields stay labelled, and no benchmark is upgraded to a measured BPC fact.

The one complete use case (used verbatim as the pitch spine): *Gomolemo’s village -> the planner’s screen -> the decision.*

3.5 What is real vs simulated

Category	Content
Real (open, hub)	Topology, generation fleet, district demand, access rates, resource layers, historical generation/import/loss series.
External (open, outside hub)	BPC loss series including 2024 (793 GWh / 16.62%); AICD/BPA 2006 grid layer; rural electrification and SAPP transfer-limit snapshots; Botswana techno-economic workbook.
Benchmarked (analogue)	Africa grid map, gridfinder, SciGRID/POMATO line parameters, PyPSA transformer/biogas/biomass/community/renewable cost tables, GEP/OnSSET method, normalized Eskom hourly shape, ENTSO-E/AEMO methods.
Assumed (labelled)	Line ratings/impedances, settlement demand allocation, demand growth, loss-load factor.
Unknown (held)	BPC ratings, switching state, substation loads, asset condition and failure history. BPC operational validation stays 0.0.

Simulated (demo)

Scenario outputs — build mixes, branch flows, criticality rankings and off-grid shortlists are model products, not measured results.

4. Implementation & Next Steps

4.1 Current status

- Problem validation completed against named open datasets (see the Problem Statement brief).
- **Four engines are implemented and running on one shared substrate:** E1 least-cost capacity screen + representative-day chronology, E2 benchmark flow/losses, E3 structural criticality, E4 VillageFit — plus an independent transmission-loss cross-check.
- **Data:** a 30-dataset catalog and a benchmark gap-resolution register; committed external datasets (filtered OSM grid + substations, AICD/BPA layer, Botswana techno-economic workbook) with SHA-256 manifests; large regional layers fetched by script.
- **Team repo:** `eukaryote1-0/grid-iq`, with CI, a PR-only workflow, tests and manifest verification.
- **Audit:** public-data planning product $\sim 9.5/10$; BPC operational validation held at **0.0**; browser E2E on the exact release still to be run on a workstation.

4.2 Limitations and mitigation

Limitation	Impact	Mitigation
Line ratings/impedances are not openly published	Constraints are estimated.	Standard-type envelopes in three cases; bracketed range; data contract for BPC.
Substation-level demand is limited (not openly published)	Demand is allocated, not measured.	Settlement/substation allocation from official context, labelled an assumption.
Failure history is not openly available	Prediction is not supportable today.	Criticality screening only; state it plainly.
Botswana-specific capex data is limited	Build-mix costs are indicative.	Botswana techno-economic workbook plus regional benchmarks, all labelled as assumptions.
No Botswana hourly load series	Chronology cannot use metered national demand.	Normalized regional shape scaled to Botswana annual/peak anchors, labelled benchmark_chronology .
OSM topology predates new lines	Missing post-2021 assets.	AICD/BPA cross-check; BPC public snapshot lists completed projects to patch.
Annual LP + representative day, not 8,760 h	Not a full production-cost model.	Position as a screening tool; the output contract says so.
Model is transmission-level; BPC losses are T&D	Direct reconciliation is approximate.	Report the transmission bracket against BPC's T&D series (642 GWh 2023; 793 GWh 2024) and explain the scope.
Topology is OSM/AICD-derived, not BPC's own	Approximate; not enough for precise power flow without ratings.	State it openly; the data contract is the path to BPC's real model.

Limitation	Impact	Mitigation
Browser E2E not verified in the build sandbox	Final visual regression unconfirmed.	Run <code>scripts/browser_smoke.sh</code> on a workstation before presenting.

Limitations we accept. We would rather name these than have a judge find them. The topology is an open approximation, not BPC’s own; line parameters are benchmarked; demand is allocated from official context; there is no Botswana hourly load series, so chronology uses a labelled analogue shape; and the distribution network is modelled only as an aggregate. None of these stop the falsifiable test we lead with — reconciling a modelled transmission loss against BPC’s published T&D series (642 GWh / 14.51% for 2023, 793 GWh / 16.62% for 2024) — and each has a named path to improvement.

4.3 Roadmap

Phase	Timeline	Activities	Success metric
1. Substrate + engines	Done	Topology, demand layers, E1–E4 on one substrate; benchmark gap-resolution register.	Four engines run on one substrate; 40 tests green; manifests verified.
2. Interface + verification	Hackathon	Wire the loss cross-check into the reconciliation panel; run browser E2E on a workstation.	A verified demo path from Overview to Criticality and VillageFit.
3. Pilot (proposed)	12 weeks post-hackathon	Reconcile with BPC; one-region screening; agree the data contract.	Deployment-ready pilot plan with partner metrics.

4.4 Partners and required resources

- **Botswana Power Corporation** — grid data, loss reconciliation, data contract.
- **Ministry of Minerals & Energy** — planning context and policy alignment.
- **BERA** — regulatory data and tariff context.
- **Statistics Botswana** — demand, census and settlement data.
- **SAPP** — interconnection capacity and regional trade options.
- **BDIH / organisers** — mentorship, government introductions, post-hackathon support.
- **In-country renewable developers** — delivery partners for whatever the siting engine recommends.
- **BIUST** — pending climate-risk and land-cover layers for siting.

5. Impact & The Ask

One anchor number. BPC’s own reports put T&D losses at **642 GWh (14.51%) in 2023** and **793 GWh (16.62%) in 2024**, independently cross-checked by the World Bank (15.35%, 2021) and AFREC (~630 GWh, 13.8%, 2023). GridIQ’s first job is to reproduce that range from open data, because a model that can reproduce the losses BPC already reports can be trusted on the decisions it recommends. (*Headline year to be fixed with the team.*)

One scale sentence. GridIQ scales by planning the existing grid as one system, with every recommendation traced to a named open dataset — no new data collection required to start.

One 12-week ask. Over 12 weeks, working with BPC and Statistics Botswana, we will reconcile the model’s loss output against BPC’s published 642/793 GWh series, complete a system-cost screening for one region’s next connection decision, and agree the data contract that turns asset-risk screening into a validated model — producing a deployment-ready pilot plan with partner-ready metrics.

6. AI Disclosure

AI tools were used to assist with research synthesis, document drafting and code scaffolding. Every data claim in this report is cited to a named source, and the team has reviewed and owns every figure and design decision. No confidential or personal data was used.